

Environmental Sensor Data Visualization Using PCA & UMAP

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INTRODUCTION

- Environmental monitoring plays an important role in understanding air quality and identifying pollution patterns. Modern environmental sensors generate large amounts of data, including temperature, humidity, PM2.5, PM10, CO, NO₂, SO₂, O₃, and other environmental parameters, making it difficult and time-consuming to analyze multiple sensor readings together. Machine learning techniques can help simplify and visualize this complex data. In this project, Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) are used to reduce multiple environmental sensor features into a two-dimensional representation. The resulting visualizations help identify similar environmental conditions, groups, and hidden patterns in the sensor data. The main objective is to compare PCA and UMAP and demonstrate how dimensionality reduction can make complex environmental sensor data easier to understand and analyze.

ABSTRACT

- Environmental monitoring systems generate large amounts of sensor data containing parameters such as temperature, humidity, particulate matter, and different air pollutants, making it difficult to directly visualize relationships and patterns among multiple variables. This project, “Environmental Sensor Data Visualization Using PCA and UMAP,” aims to simplify and visualize environmental sensor data using Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP). The selected environmental features are preprocessed, standardized, and transformed into a two-dimensional representation while preserving important information from the original dataset. The results are visualized using 2D scatter plots, where similar environmental observations appear closer together, helping to identify environmental patterns, similarities, and groups while comparing PCA and UMAP..

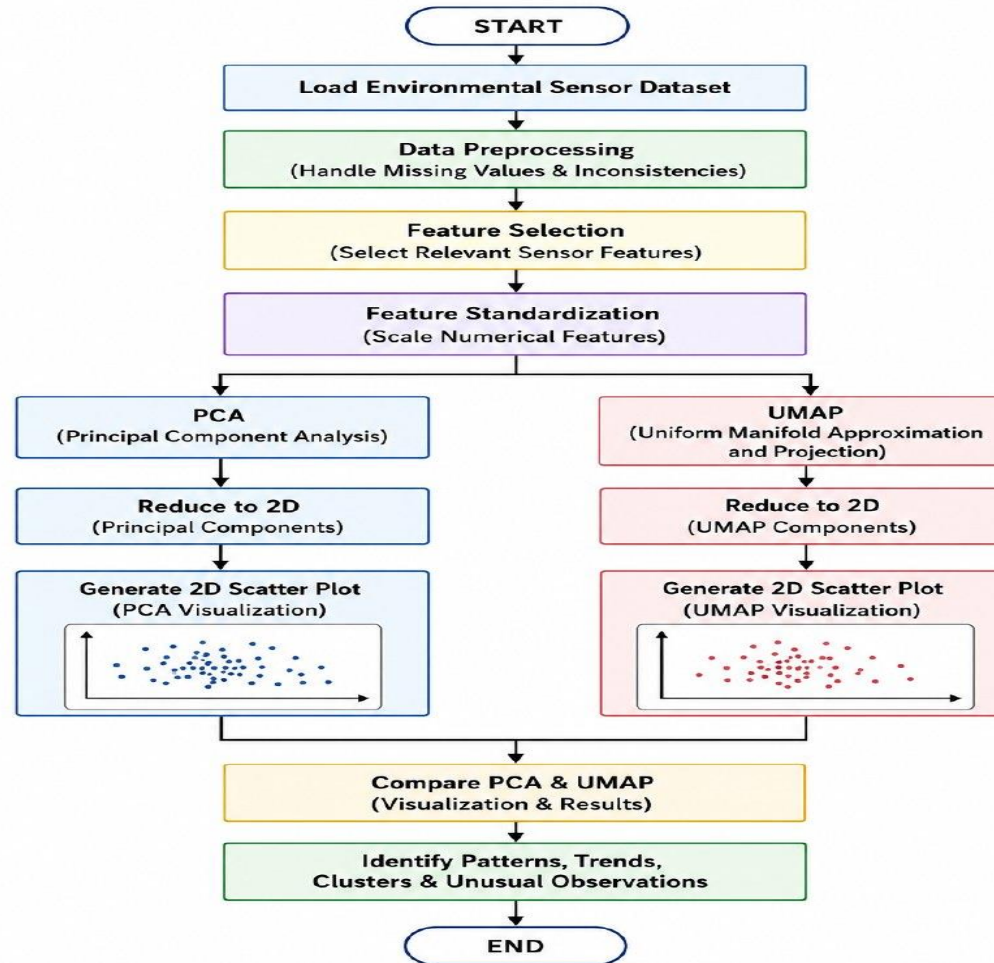
EXISTING SYSTEM

Traditional environmental data analysis mainly uses tables, basic charts, and individual sensor comparisons. When multiple parameters such as temperature, humidity, PM2.5, PM10, and other pollutants are analyzed together, it becomes difficult to understand the relationships and hidden patterns in the data. PCA and UMAP can help reduce and visualize high-dimensional data, but the clusters formed by UMAP can be difficult to interpret using the original environmental variables. Therefore, additional analysis is needed to understand the characteristics of each cluster.

PROPOSED SYSTEM

The proposed system uses PCA and UMAP to reduce and visualize complex environmental sensor data. The data is first preprocessed and standardized, followed by dimensionality reduction and 2D visualization. Along with displaying the clusters, the system performs cluster-wise analysis of the original environmental variables and explains the characteristics of each cluster. For example, it can show whether a cluster has higher PM2.5, higher temperature, or lower humidity. This makes the clusters easier to understand instead of only showing them visually.

FLOW DIAGRAM



ALGORITHM

- Load the environmental sensor dataset.
- Preprocess the data by handling missing values and selecting the required environmental sensor features.
- Standardize the selected numerical features so that all sensor parameters are on a comparable scale.
- Apply PCA (Principal Component Analysis) to reduce the high-dimensional environmental data into two principal components.
- Apply UMAP (Uniform Manifold Approximation and Projection) to reduce the same environmental data into two dimensions.
- Generate 2D scatter plots for both PCA and UMAP to visualize relationships between environmental observations.
- Compare the PCA and UMAP visualizations to identify similar environmental conditions, groups, and patterns.
- Analyze the results and determine how effectively PCA and UMAP represent the environmental sensor data.

EXPECTED OUTPUT

- The project is expected to generate 2D visualizations of environmental sensor data using PCA and UMAP. PCA Visualization: Helps show the overall variation in sensor data by reducing multiple features into a simple 2D representation. UMAP Visualization: Helps show how similar environmental sensor readings are related and grouped together. Sensor readings with similar environmental conditions are expected to appear closer together in the visualizations. The results will help users easily understand patterns, trends, and clusters in the sensor data. Comparing PCA and UMAP will help understand how each technique represents the same environmental data differently.

THANKYOU